

class10

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PDB Statistics

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

93.7892% of structures in the PDB are solved by X-Ray and EM.

```
# Importing PDB file
pdb_stats <- read.csv("pdb_stats.csv")

# Percentage of structures solved by X-ray and EM
((sum(pdb_stats$X.ray) + sum(pdb_stats$EM))/sum(pdb_stats$Total))*100
```

```
[1] 93.7892
```

Q2: What proportion of structures in the PDB are protein?

97.9118% of structures in the PDB are protein.

```
# Proportion of structures that are protein
total_protein <- pdb_stats$Total[1:3]
(sum(total_protein)/sum(pdb_stats$Total))*100
```

```
[1] 97.9118
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1,227 HIV-1 protease structures in the current PDB.

Visualizing the HIV-1 protease structure

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Only the oxygen molecule is shown.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

HOH308

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

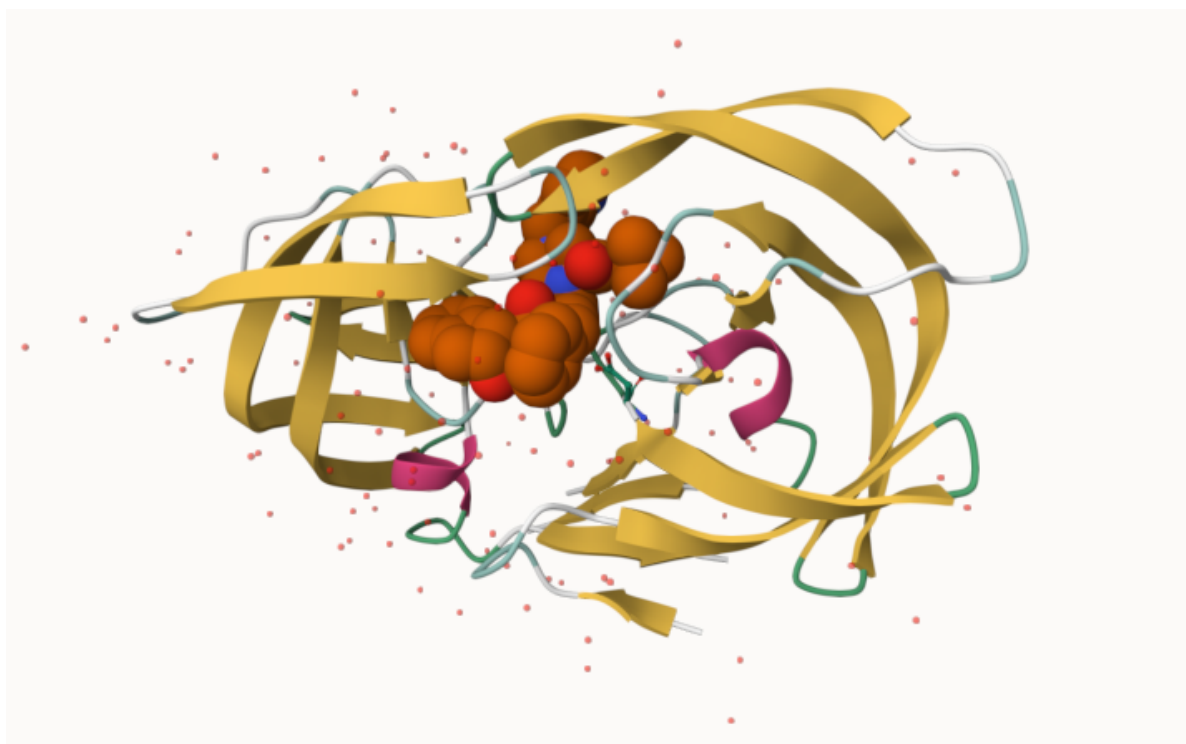


Figure 1: 1HSG

Introduction to Bio3D in R

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198 amino acid residues.

Q8: Name one of the two non-protein residues?

Nucleic atoms.

Q9: How many protein chains are in this structure?

There are two protein chains.

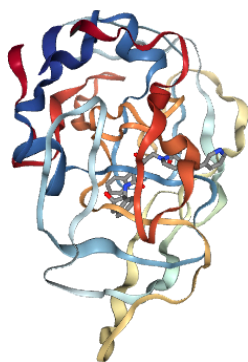
```
# PDB attributes
attributes(pdb)
```

```
$names
[1] "atom"    "xyz"     "seqres"  "helix"   "sheet"   "calpha"  "remark"  "call"
```

```
$class
[1] "pdb" "sse"
```

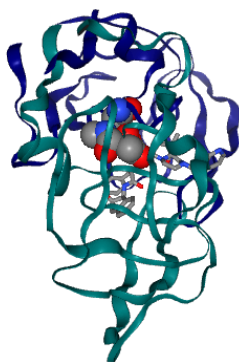
```
library(bio3dview)
library(NGLVieweR)

# Structure overview of pdb
view.pdb(pdb) |>
  setSpin()
```



```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy","teal"),
         highlight = sele,
         highlight.style = "spacefill") |>
setRock()
```



Predicting functional motions of a single structure

```
# Loading Adenylate Kinase structure
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

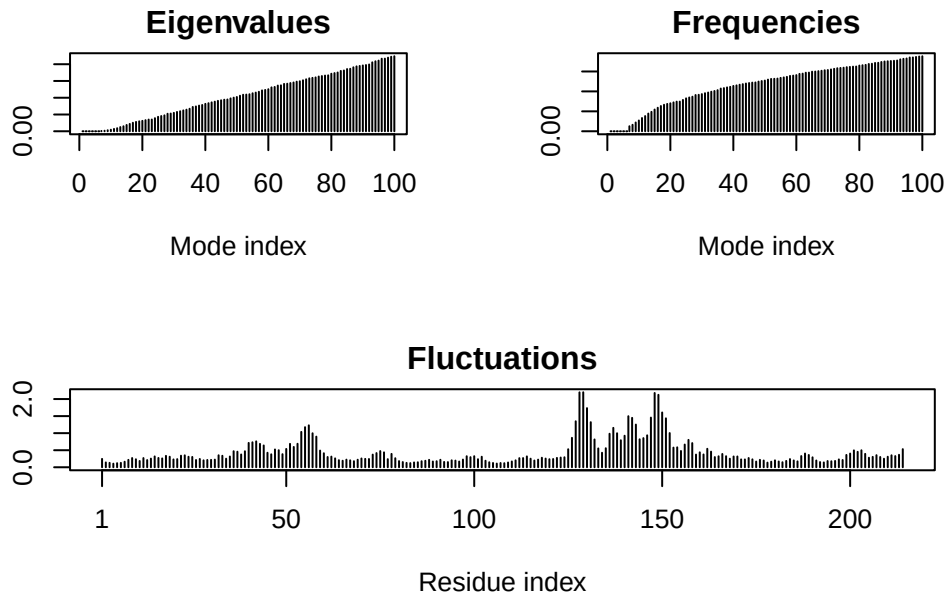
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
# Performing flexibility prediction
m <- nma(adk)
```

```
Building Hessian... Done in 0.025 seconds.
```

```
Diagonalizing Hessian... Done in 0.352 seconds.
```

```
plot(m)
```



```
mktrj(m, file="adk_m7.pdb")
```

Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

The “msa” package.

Q11. Which of the above packages is not found on BioConductor or CRAN?

The “bioboot/bio3dview” package.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True.

```
library(bio3d)  
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRPTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

     121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTPALIG
     121      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214 amino acids

```
hits <- NULL
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6H
```

```
# Download related PDB files
```

```
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3HPR.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4V.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/5EJE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4Y.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3X2S.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6HAP.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6HAM.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/4K46.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3GMT.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download
```

		0%
=====		8%
=====		15%
=====		23%
=====		31%
=====		38%
=====		46%
=====		54%
=====		62%
=====		69%
=====		77%
=====		85%
=====		92%
=====		100%

```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
```

```

pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

```

Extracting sequences

```

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdbs/split_chain/6HAM_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbs/split_chain/4K46_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdbs/split_chain/4PZL_A.pdb

```

Annotate collected PDB structures

```
# Vector containing PDB database codes
ids <- basename.pdb(pdb$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
# Viewing available annotation data
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					
1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					

4K46_A	4K46	A	Protein	214	X-
ray					
3GMT_A	3GMT	A	Protein	230	X-
ray					
4PZL_A	4PZL	A	Protein	242	X-
ray					

	resolution	scopDomain	pfam
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
3X2S_A	2.80	<NA>	<NA>
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)
3GMT_A	2.10	<NA>	<NA>
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)

	ligandId
1AKE_A	AP5
6S36_A	CL (3),NA,MG (2)
6RZE_A	NA (3),CL (2)
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,CO
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG
6HAP_A	AP5
6HAM_A	AP5
4K46_A	ADP,AMP,PO4
3GMT_A	SO4 (2)
4PZL_A	CA,FMT,GOL

	ligandName
1AKE_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6S36_A	CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A	SODIUM ION (3),CHLORIDE ION (2)
3HPR_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
1E4V_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	

5EJE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6HAM_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
4K46_A	ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A	SULFATE ION (2)
4PZL_A	CALCIUM ION,FORMIC ACID,GLYCEROL

source

1AKE_A	Escherichia coli
6S36_A	Escherichia coli
6RZE_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
3GMT_A	Burkholderia pseudomallei 1710b
4PZL_A	Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBIT

6S36_A
6RZE_A
3HPR_A
1E4V_A

loop

5EJE_A

1E4Y_A

loop

3X2S_A

conjugated adenylate kinase

6HAP_A

6HAM_A

4K46_A

3GMT_A

4PZL_A

Cryst

The crys

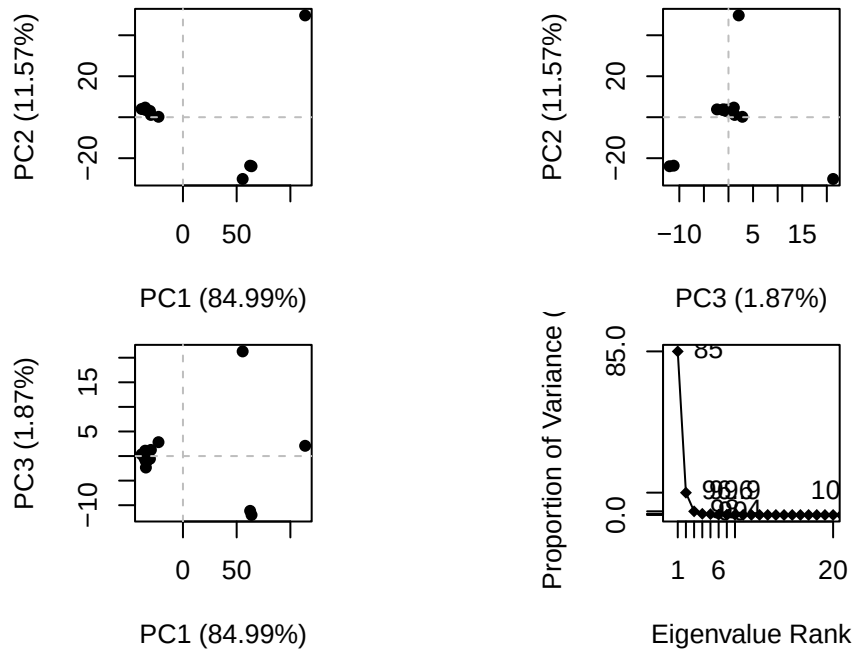
citation rObserved rFree

1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560
6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al. Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al. To be published	0.19360	0.23680

	rWork	spaceGroup
1AKE_A	0.19600	P 21 2 21
6S36_A	0.15940	C 1 2 1
6RZE_A	0.18190	C 1 2 1
3HPR_A	0.20620	P 21 21 2
1E4V_A	0.19600	P 21 2 21
5EJE_A	0.18630	P 21 2 21
1E4Y_A	0.17800	P 1 21 1
3X2S_A	0.20700	P 21 21 21
6HAP_A	0.22370	I 2 2 2
6HAM_A	0.20311	P 43
4K46_A	0.16730	P 21 21 21
3GMT_A	0.23500	P 1 21 1
4PZL_A	0.19130	P 32

Principal component analysis

```
# Perform PCA
pc.xray <- pca(pdbx)
plot(pc.xray)
```

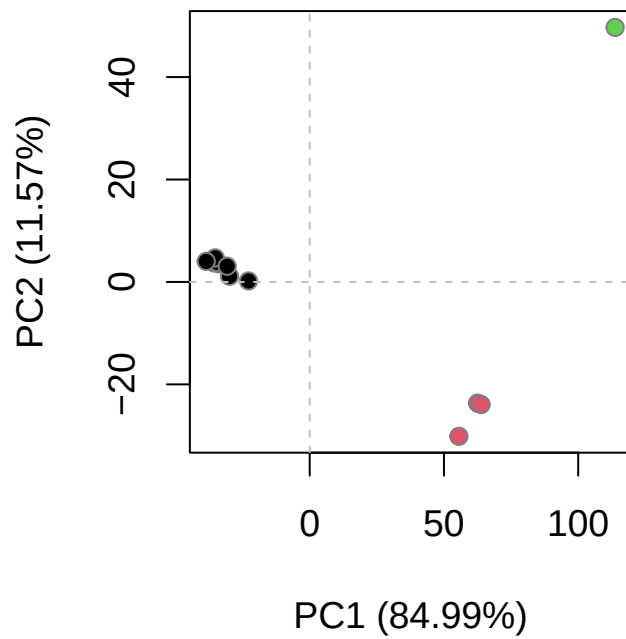


```
# Calculate RMSD
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



PCA visualization

```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```

